Agrivoltaics: the dual-use solution for agriculture and renewable energy in Africa

Introduction

Africa stands at the crossroads today with the interrelated challenges of food vulnerability, ineffective energy access, and increasing exposure to climate change. These are not singular but are rather interlinked in a tight web of interlinkages that call for multifaceted, interlinked system responses. In response to mounting challenges, agrivoltaics, a new paradigm of integrated solar photovoltaic (PV) power generation and livestock or crop farming on the same surface area, has been emerging as a possible double-use system since it can solve the food-energy-water nexus problem at once (Barron-Gafford et al., 2019).

Agrivoltaics was first proposed by Goetzberger and Zastrow in 1981 as a concept for the two-use of land for energy crops and solar panels. This came as a result of the authors' anticipation that conflict of land between agriculture and energy crops would rise with the possibility of an increasing world population and rising energy demands. While it was an intellectual construct initially, the concept has matured into an applicable solution with practical application on over a continent, with its pilot implementations on the African continent. Its applicability is particularly pertinent to Africa, with over 600 million Africans without access to electricity and a good number more depending on smallholder farming for sustenance and livelihood.

Agrivoltaics improves land use effectiveness and rural socio-economic growth, renewable energy yield, and farm sustainability. Strategic co-siting of crop and PV array addresses the adverse effects of climate change by reducing soil water evaporation loss and crop heat stress, the two major risks for agriculture in arid and semi-arid environments. In addition to this, it also generates clean energy that could be used for irrigation, storage, lighting, and even for generating revenues as an additional value by being supplied to the grid or microgrids in the area (U.S. Department of Energy, 2022).

Other than this, agrivoltaics is already heading in the right direction with the broad vision for sustainable development, particularly per the United Nations Sustainable Development Goals (SDGs) like SDG 2 (Zero Hunger), SDG 7 (Affordable and Clean Energy), and SDG 13 (Climate Action). In facilitating rural communities to produce their renewable energy without compromising food agriculture, agrivoltaics facilitates bridging development backlogs that have long constrained development at the periphery of Africa. Its adoption and scaling, however, are subject to general understanding of the environment, community engagement, technology transfer, and progressive policy landscapes (Randle-Boggis et al., 2024). It aims to provide a synoptic analysis of its historical development, new technology design, socio-behavioral consequences, and contextual appropriateness for various African agro-ecological zones. The research also provides comparisons of available alternatives across various contexts and explores long-term socio-environmental consequences, desirable and undesirable. As one can observe from this analysis, agrivoltaics is not merely a

technology intervention and is rather a paradigm shift that rethinks the possibilities of how Africa may pursue bottom-up sustainability.

Historical development of Agrovoltaic

The history of agrivoltaics charts a course of development over decades from concept to an integrated strategy towards solutions of pressing global sustainability challenges. Adolf Goetzberger and Armin Zastrow coined the term "agrivoltaics" in 1981 in a visionary paper on the use of land for both solar power generation and food crops. They cleared the way for an entirely new paradigm of land use planning and renewable energy integration by embracing the fact that the world's increasing demands for energy and food would necessarily place land at odds, especially in regions of scant resources such as sub-Saharan Africa (Barron-Gafford et al., 2019). Theory was revolutionary; however, early attempts were simplistic and far too frequently unrealistic.

The earliest agrivoltaic installations during the 1980s and 1990s were by individual ground-mounted solar panels of low roof height. The initial designs were, in many ways, wanting. The panels cast too much shade on crops beneath, significantly reducing the light supply to photosynthesis. The low clearance height of modules from the ground also prohibited the use of farm machinery, leading to wasteful and labor-intensive farming (Barron-Gafford et al., 2019). These logistical constraints brutally limited the scalability of agrivoltaic systems to pilot or proof-of-concept projects for decades.

A technological innovation began unfolding from the early 2000s, particularly in countries such as Germany and Japan. Germany, with the Energiewende (energy transition) policy encouraging expenditure on renewable technology, saw scientists begin implementing solutions to lift solar panel frames high enough for farm machinery to drive underneath and reduce shading of crops. They employed high-density PV panel configurations, and along with these, utilized wide panel spacing and tilt angle adjustable mounts to allow increased dispersal of the sun over the canopy and on the ground.. Japan also created modular agrivoltaic systems for small-scale farming, featuring flexible support systems that respect the form of conventional farming systems. These innovations paved the way for agrivoltaics as a research specialty, a big picture field of study, and not an idea. New forces joined the research and development scene of agrivoltaic systems from nations like France and the United States in the 2010s.

Energy generation as well as crop generation began to be optimized here by innovation. Solar tracking systems that can change panel angles throughout the day were installed more frequently, such that dynamic shading regimes could adjust based on evolving crop requirements (Willockx et al., 2023). Computer sensors to track environmental parameters such as temperature, solar radiation, and soil moisture also began to play a crucial part in managing system operation. More evidence at higher volumes through these deployments of agrivoltaics has recorded positive outcomes. It proved the reality that certain crops were positively affected by improved microclimate conditions and reduced water stress (PLOS,

2018). The most dramatic shift of agrivoltaics has indeed taken place in the 2020s with the incorporation of newer technologies such as artificial intelligence (AI) and the Internet of Things (IoT).

Computer simulations can now handle real-time sensor readings from the environment to better optimize panel tilt, irrigate with greater precision, and even predict crop health through thermal and multispectral imaging (U.S. Department of Energy, 2022). It all adds a high level of responsiveness and efficiency to agrivoltaic management, one that is even more desirable in regions plagued by unpredictable climatic events, i.e., the Sahel and the Horn of Africa. For instance, artificial intelligence-based panel rotation systems reduce energy yield maximization while simultaneously reducing heat stress on crops through intensity and duration management of shading (Zotti et al., 2024). Agrivoltaic technology gradually begins to catch on in Africa as pilot studies and feasibility tests begin to make the argument for the viability of the technology under diverse agro-ecological conditions.

Kenyan, South African, and Ghanaian feasibility studies have demonstrated the viability of current agrivoltaic technologies to be integrated with maize, beans, and horticultural produce (Randle-Boggis et al., 2024). The schemes also illustrate how community schemes, in addition to adequate training and financing models, can overcome the adoption hurdle at a nascent stage. Since technology is more modular, longer-lasting, and more compatible to interface into African agricultural systems, the difference between applications of innovation is narrowing.

Design Evolution and Capabilities

The agrivoltaic system design has also changed to achieve the optimal fine compromise between sustainable agriculture and maximum photovoltaic (PV) efficiency. There is a progression that ensues in a bid to achieve the solution to the inherent compromise between the shading effect of the solar panel and the need for light by the crops. Previous agrivoltaic design sacrificed such balance, prioritizing electricity generation over crop production (Barron-Gafford et al., 2019). With time and as the topic evolved, interdisciplinarity ensured that there was a revisiting of design strategy to ensure food and energy production activity compatibility.

One of the first design improvements was to raise the solar panels, but one that allowed sunlight onto crops and not over them, and allowed mechanized farm tools to travel underneath. High-clearance frames, hung usually 2–4 meters above the earth, allowed mechanized farming to continue unchecked (Campana et al., 2021). This was especially useful for intensive farm production when labor-saving machinery such as combine harvesters and tractors is needed. Besides, greater inter-row space allowed diffused light to pass through inter-row crops, enhance photosynthesis, and limit yield loss. Such skyscrapers, however, enhanced material expense and wind load susceptibility, thus leading to a trade-off in cost-effectiveness and physical strength (Randle-Boggis et al., 2024). Finally, adaptive solar tracking systems were the most advanced innovation in agrivoltaic design. Compared to

fixed-tilt panels, the systems track PV modules during the day to tilt in the direction of the sun to collect maximum solar energy.

Surprisingly, tracking systems may be programmed with dynamic shading level control based on plant growth stages and climatic conditions (Willockx et al., 2023). An icon is employed to represent a company or an organization. For example, even on the hottest midday sun, panels can be fitted to shade partially and thus prevent evapotranspiration and also shield crops against heat stress. Intellishading is very effective with lettuce, spinach, and some legumes and results in biomass yield and water-use efficiency (PLOS, 2018). Integration has further been expanded to AI-based decision-making systems and IoT-based sensors. They constantly monitor soil water levels, ambient humidity, temperature, solar irradiance, and plant health indicators. AI algorithms' real-time analysis offers recommendations or even executes them independently for panel tilt, irrigation planning, and pesticide spraying (Zotti et al., 2024). Such precision agriculture PV infrastructure is playing a major role in the optimum use of resources and generation of additional energy and food output. Besides, there is a queue of future long-term structural and technology problems to face agrivoltaic systems.

The planning, for example, of the best vineyard layout in the southern part of France would be inappropriate for an open-field maize farm in Kenya or a rice paddy in Southeast Asia. These encompass cloud cover, temperature gradient, and rainfall pattern that are highly heterogeneous across regions in a way that region-specific engineering and agronomic design will be necessary (Randle-Boggis et al., 2024). Other than that, the selection of crops for agrivoltaic layouts is also constrained since most of the cereal crops like wheat and maize are high-intensity light-demanding and less tolerant to the prevailing partial shading conditions in agrivoltaic layouts. Even the operation and maintenance are a challenge. Solar trackers and arrays of sensors need to be calibrated and shielded at regular intervals from dust, insects, and mechanical stress. Maintenance requirements in the majority of African rural environments may restrain large-scale agrivoltaic installation work, where expert knowledge and spares are readily available (U.S. Department of Energy, 2022).

In addition, initial capital investment in installing elevated, tracking-able, and sensor-based systems is a poor smallholder farmer's biggest barrier when there is no genuine external assistance. In terms of energy intermittency, a technological imperative exists. Solar power generation is day-night and weather dependent, and therefore, individual agrivoltaic systems are unreliable unless onboard energy storage technology is employed. Battery systems are effective but costly and environmentally expensive, especially when lithium-ion technology is used. Other technologies for energy storage, such as hydrogen gas generation and thermal storage systems, are other alternatives, but at the moment, only at the pilot or experimental level (Zotti et al., 2024).

Technical issues, including specificity of design, maintenance complexity, costly initial rollout, and institutional barriers, remain to undermine large-scale rollout, especially in the Global South. The issues need to be addressed by technological advancement and

participatory policymaking, building capacity, and local research to make agrivoltaic systems economically viable and sustainable across the varied agricultural contexts.

Benefits and Limitations of Agrovoltaics

Agrivoltaics is such a beautiful association of food and energy production along with climatic resilience opportunities, agrosustainability, and rural power provision. Combining farm crops and photovoltaic panels in a single land use, agrivoltaics aims to solve simultaneously the urgent challenges of land shortage, energy access, and climatic resilience. But since such double advantages do exist, the technology is also hindered by a variety of technical, socio-economic, and environmental constraints that need to be addressed if it is to be transformed into an inclusive and scalable option. Perhaps one of the greatest advantages of agrivoltaics is the ability of the technology to enhance land-use efficiency by making double land functionality.

Where land competition is intense or cultivated land is scarce, e.g., Sub-Saharan Africa and Southeast Asia, food and energy are combined, and thus land is not devoted to one end use at the cost of the other. Dual-use land systems reflect an LER value higher than one, i.e., crop and electricity productivity at the same time is greater than individual plants, according to Barron-Gafford et al. (2019). It is thus because it is especially well-adapted where energy poverty and food insecurity exist. Agrivoltaic systems also aesthetically change the microclimate.

Sun panel shading reduces crops' exposure to excess heat and sunlight, both of which are harmful to yields. Willockx et al. (2023) have established that sun panel shading inhibits evapotranspiration rates, which leads to increased soil water retention capacity and reduced irrigation needs. Such water saving is advantageous in hot and arid climatic situations where water stress is a growth-limiting factor for plants. Microclimate beneath the panel also lengthens the growing season of some crops and gives the ideal condition for value-added, shade-bearing crops such as lettuce, herbs, and tomatoes (PLOS, 2018). At the power level, agrivoltaics produce clean, green energy to energize diversified farming, such as processing, cold storage, and irrigation.

Decentralization of solar power renders power systems off-grid, low-cost, and less reliant on high-cost or unpredictable national grids. especially in rural, off-grid, and remote locations where extension of the grid is not economically viable, Zotti et al. (2024) observe that agrivoltaics can guarantee 24-hour power access by battery storage or mini-grids for household and small-scale agro-industry uses. Agrivoltaics ensures farms become profitable economically by minimizing the cost of energy expenditures and potential sales of surplus electricity to the grid or local users. This diversification of revenue protects farmers from change loss of crops and unstable market prices. Agrivoltaic farms secondly provide rural employment in the repair, maintenance, and inspection of solar panels, thereby contributing to rural economic development (Randle-Boggis et al., 2024).

Among its most serious shortcomings is the prohibitively high initial capital investment in agrivoltaic facilities, namely in the use of advanced technologies like solar trackers, IoT sensors, and smart irrigation systems. The question of cost is therefore so essential in developing nations where farmers do not have access to credit services such as subsidization or specialized financing structures to invest in renewable energy (U.S. Department of Energy, 2022). Besides that, the majority of accessible agrivoltaic empirical experiences are based on pilot tests that have been carried out in high-income economies, and their business model will not be easily translatable for direct application to African smallholder agriculture systems, Asian systems, or Latin American systems. Technical complexities in operating dual-use systems are the second constraint.

They demand constant, around-the-clock monitoring and careful engineering to accurately enhance plant light quality and PV efficiency adequately. Even slight panel misalignment or inadequate control over crop selection can lead to an optimal energy and agriculture yield.

Crop staples intolerant of shade, like rice, wheat, and maize, are impacted by reduced yield with partial shading, constraining agrivoltaics size to smaller quantities of special or value crops. This limits the scalability of agrivoltaics to higher levels of food shortage in regions reliant upon such crops (Randle-Boggis et al., 2024). Farm activities can also be disrupted by panel installations. For instance, overhead installations can obstruct gargantuan farm machinery or make activities such as harvesting and weeding more complex. There are also environmental issues related to the material used in PV systems, particularly in mass roll-out. Land disturbance during installation, manufacturing the carbon footprint of panels, and solar panel end-of-life disposal are some of these. Although agrivoltaics increase renewable power, long-term sustainability is made dependent on circular design and green waste treatment systems, which are usually underdeveloped or inefficiently deployed (Zotti et al., 2024).

Lastly, agrivoltaics offer a fresh opportunity to address the world's several problems concerning food security, supply of clean energy, and climate resilience simultaneously. Its benefits ranging from more productive land through enhanced water and energy efficiency are vast and well established.

But to realize its potential, it needs to overcome a set of challenges whose large consists of implementation costs, technological innovation, growability, and socio-political resistance. Overcoming all of them will necessarily require collective effort by research, policy, finance, and stakeholder action on conditions that make agrivoltaic technologies inclusive, resilient, and sustainable in the various contexts to which they are most stretched.

Social-economic and environmental Impacts of Agrivoltaics

Agrivoltaics stretches itself as a cutting-edge convergence technology of clean energy and sustainable agriculture, whose extensive socio-economic and environmental impacts must be examined critically. Putting photovoltaic (PV) electricity production on the same ground area as that of agriculture, agrivoltaic systems attempt to prevent the triple curse of climate

exposure, food poverty, and energy poverty. In the absence of policy support with inclusiveness and contextualization, however, it can work as a mechanism of further deepening prevailing socio-economic disparities and triggering unintended ecological consequences.

Socio-economically, agrivoltaics hold humongous promise for rural communities to earn diversified revenues. Agrivoltaics subject the farmer to humongous climatic and market risk, with production at the mercy of capricious weather, pests, and fluctuating market prices. With farmers putting up the solar panels, ancillary income comes from selling surplus power to the grid or utilization in rendering value-added farm services such as irrigation, milling, or cooling. This type of income diversification, as noted by Randle-Boggis et al. (2024), enhances economic resilience in off-grid or unelectrified smallholder farmers. Early pilot agrivoltaic experiments, for example, started in Kenya with support from local cooperatives and have been shown to double household incomes by 30% through diversified incomes from farm sales and selling electricity (Willockx et al., 2023).

Not only does it start earning revenue instantaneously, but agrivoltaics also creates employment in the clean energy value chain. Installation, maintenance, monitoring, and training of farm operations under agrivoltaics bring new avenues to employment for young people and women who otherwise have no stake in the traditional farm labor force. Zotti et al. (2024) assume that rural solar job programs have been attributed to Indian agricultural production, employing thousands of individuals through localized solar production and social entrepreneurship. Moreover, with cooperative and microgrid assistance, agrivoltaics can even make rural consumers into producers and consumers of electricity and hence open the door to energy democracies and community ownership schemes.

Agrivoltaics also plays a commercial and significant role at the regional and local levels of energy access. The grid facilities of most of the developing countries do not exist or are in short supply in villages. Decentralized solar power can reduce the cost of energy and enhance access to basic services such as health, education, and agricultural processing. Health conditions and education of individuals living in areas where on-site solar power is available are improved by lighting, refrigeration, and communication, noted the U.S. Department of Energy (2022). The social benefits fall most broadly to women and children, those who are most at risk of energy poverty.

It is economically beneficial as agrivoltaics promotes land saving and diversification. Agrivoltaics reduces forest or marginal land tension that needs to be cleared before planting solar farms or farms by enabling two squares of land to be utilized. Land-sparing occurs every time land use conflict is the final driver of environmental loss. Barron-Gafford et al. (2019) had previously shown that agrivoltaic systems reduce the amount of land disturbance, thereby reducing land loss as well as land erosion. Shading panels also increase water storage and reduce irrigation consumption, as observed in dry regions like Arizona and much of East Africa, where water is greatly needed (PLOS, 2018).

On top of that, crops grown under medium shading are less reliant on chemical inputs such as pesticides and fertilizers because the modified microclimate inhibits pest infestation and slows nutrient release in the soil (Randle-Boggis et al., 2024). Such synergies in nature place agrivoltaics on a very promising path of both adaptation to and mitigation of climate at the same time, especially under the auspices of national energy or agriculture policy reform.

Its socio-economic impact is not equal in proportion. Technology access, land access, and access to money disproportionately impact societies. The enormous initial investment of having the systems in place can even lead to perpetuation of inequality if the system access is only for affluent farmers or business groups. Without inclusive finance systems such as subsidies, microcredit, or public-private partnership arrangements, poor farmers and poor smallholder farmers will be systematically excluded from the agrivoltaic revolution. Zotti et al. (2024) also posit that without community participation, agrivoltics can potentially cause land tenure conflicts or disrupt customary patterned agriculture, particularly if the land is communally or Indigenous owned.

All that is achieved is accompanied by watchful government regulation of finance, land ownership, and nature. Agrivoltaics cannot be replicated in template form but must be adapted to local contexts in participatory planning, reasonableness of science, and ordinary people's participation. It can do so on its own in such a way as an attempt to play its part as a spark to initiate change towards a green and equitable future.

Case studies of Agrovoltaics in different Countries

Agrivoltaics evolved rapidly from theory to practice, being utilized in other countries with different climatic, economic, and social conditions. Because the countries were experiencing a climate crisis and food unavailability, the agrivoltaic systems were piloted and scaled up in countries with proven feedback. Each one of these India, Kenya, Japan, and the United States level case studies shows the potential of agrivoltaics for advancing land use efficiency, farmers' incomes, and renewable energy objectives. Each lesson is, nevertheless, tempered by some difficulty and learning, and they work to provide visibility to the merits of locally adapted substitutes alongside participatory governance institutions.

Agrivoltaics has been identified as a major part of India's rural development and renewable energy progress national programme. The government of India has promoted the deployment of solar panels on farms as part of the KUSUM (Kisan Urja Suraksha evam Utthaan Mahabhiyan) programme to allow farmers to generate electricity for irrigation and sell surplus power to the grid (Zotti et al., 2024). The program has made the farmers self-sufficient by reducing their reliance on expensive diesel pumps and maximizing farming yields through electrification. For instance, Maharashtra agrivoltaic farms and the semi-arid areas of Gujarat have proved that tomatoes, onions, and spinach grow best under canopy shade from solar panels, which gives farmers two incomes (Randle-Boggis et al., 2024). But there are challenges, particularly in areas with poor grid infrastructure, a lack of maintenance competence, and a bureaucratic process of withholding disbursement of subsidies. But Indian

experience provides policy-driven agrivoltaics as an antidote to root out energy poverty and farm distress.

Agrivoltaics tailored to local socio-economic conditions is another such strong example from Kenya. Agrivoltaic projects in this East African nation with a high rural electrification gap and frequent droughts have been wonderful feats towards sustainable development. Pilot projects funded by donor NGOs and research organizations have proven the capability of solar panels to improve water-use efficiency through reduction in evaporation and generating clean energy for driving irrigation pumps (Willockx et al., 2023). The systems tend to be community-managed under a cooperative mode dominated by women, fostering gender empowerment and social inclusiveness. Shading by the PV modules enhances productivity and extends the growing season for heat-sensitive crops (Randle-Boggis et al., 2024). Though affordability is a key limiting factor, the upfront installation costs are outside of the reach of the majority of smallholder farmers in the absence of external intervention. Moreover, traditional land management systems sometimes clash with agrivoltaics' strict infrastructure, which requires diplomatic engagement and participatory planning.

In Japan, where land is in short supply and agricultural land comes into conflict with urban and industrial development, agrivoltaics, or locally known as "solar sharing," has been continuously increasing in popularity as a means to resolve land use conflicts. Japanese agrivoltaic farms usually elevate solar panels above average rice paddies or vegetable farms in a manner such that crops receive solar light while electricity is generated on the surface (Barron-Gafford et al., 2019). They are invariably combined with smart farm systems that make use of sensors and automation to optimize water, light, and nutrient inputs. They are supported by the Japanese government through regulatory reforms and feed-in tariffs, which make solar energy production economically feasible for farmers to pursue. A case in Chiba Prefecture illustrates that farmers who were growing lettuce and ginger under PV panels had healthy crops with reduced water usage, and a new source of income from the sales of electricity (Randle-Boggis et al., 2024). Critics would counter, though, that a focus on energy generation will take attention away from issues in food production, especially when investors are less concerned with upholding local food security than they are with profit. The Japanese experience, therefore, requires finding a balance between agricultural and food objectives in agrivoltaic systems.

Agrivoltaics is gaining momentum in the United States, with Arizona, Colorado, and Massachusetts taking the lead, fueled by experimentation due to innovation in university research, government support, and farm patronage. Agrivoltaics has been shown by experiments at the University of Arizona to make crops more climate-resistant at the cost of lower water demands a huge advantage for America's Southwest (Barron-Gafford et al., 2019). Massachusetts' SMART state program offers economic incentives for dual-use solar farms, encouraging pasture, hay, and vegetable crops to be blended with PV systems. Land use planning and interconnection issues with utilities have sometimes slowed overall adoption of agrivoltaics. On the other hand, these international case studies show that while agrivoltaics offers an all-encompassing package of benefits land optimization, income

diversification, power access, and ecological sustainability the success is extremely site-specific.

Climate, crop type, tenure system, cultural setting, and government support are some of the main success determinants. Agrivoltaics has high potential for poverty alleviation in Kenya and India if supplemented with community and capacity-building approaches. Except for experience in Japan and the U.S., they emphasize the need for technology harmonization, careful regulation, and concordance with high-tech farming. The general moral that may be gleaned is clear: context-aware planning, inclusive deployment, and ongoing institutional support are what make agrivoltaics become all it may be. Briefly, agrivoltaics are not a typical innovation but a model that can be made context-flexible for agro-ecological and socio-economic conditions. The case studies suggest the revolutionary and banal experience of scale-up of agrivoltaic systems around the world. With rising climate change and intensifying land-use pressure, strategic siting of agrivoltaics by being sensitive to local contexts and facilitated through participatory governance can be one of the rural sustainable development as well as world climate resilience.

User Behavior and Social Dynamics

The performance of agrivoltaic systems also does not solely rely on the performance of the technology but on the intricate mix of user behavior, cultural practices, and social processes. In rural societies in general, and those of Africa and South Asia in particular, the uptake of agrivoltaic systems primarily depends on technology compatibility perception with the existing farming customs and society participation levels in planning and decision-making activities. Evidence indicates that when local stakeholders are involved directly in system design and governance, the likelihood of adoption success increases (Randle-Boggis et al., 2024; Willockx et al., 2023).

Behavioral economics provides the necessary explanations of why agrivoltaics will be adopted by some communities and not others. Adoption is influenced by psychological factors of loss aversion, status quo bias, and future discounting. For instance, although long-term gains can be seen, farmers may fail to adopt new, untested technologies due to short-run risks of disrupting production or income (Zotti et al., 2024). Secondly, social proof and normative influence learning from others who have successfully implemented new practices may be powerful drivers. In Kenya, as an example, decision-making via cooperatives yields higher levels of adoption of agrivoltaic systems where innovators have a chance to demonstrate provable benefits (Willockx et al., 2023).

Knowledge and information exchange are also critical. Denying one access to quality information about agrivoltaics will lead to misinformation and underutilization. In India, solar take-up in agriculture has been subsidised heavily under the KUSUM scheme, but take-up is widely scattered, partly due to ignorance and lack of training infrastructure (Zotti et al., 2024). In Japan, however, promotion and embedding of education in agricultural

extension services have achieved cultural fit of the innovation with extant values like land stewardship and cooperation (Randle-Boggis et al., 2024).

Regional Relevance: African Context

Africa's unique socio-economic and environmental context is, in particular, a crisis environment into which agrivoltaic systems can be introduced. Sub-Saharan Africa is a region of high food insecurity, low access to electricity, especially rural access, and increasing vulnerability to climate change. More than 600 million Africans lack access to electricity, the International Energy Agency states (IEA, 2022). At the same time, nearly 33 million African smallholder farms produce nearly 70% of the continent's food, thereby more prominently highlighting the importance of agriculture to livelihood and economic wealth (FAO, 2021).

Agrivoltaics does so in a single step by allowing individuals to colocate food and energy production at the same time, improve land-use efficiency, as well as climate resilience. Agrivoltaics have also been proven in Kenya on-farm tests to increase spinach and tomato yields by reducing heat stress and evapotranspiration (Willockx et al., 2023). Local power generation also reduces diesel generator use and allows for extra irrigation, cold storage, and processing capability life-giving services not easily accessible in most off-grid areas (Randle-Boggis et al., 2024).

But limitations of the African climate as well must be overcome before roll-out is possible in numbers. Uncertainty of land ownership, lack of capital, and lack of willingness by the government to help are some of the barriers limiting roll-out in numbers. For instance, all of the agriculture takes place on communally owned, and it would not be easy to put solar panels there. Farmers will not take up long-term opportunities such as these unless they are assured secure land tenure and policy-friendly interventions (Zotti et al., 2024). Gender equality must also be dealt with: women constitute the biggest community of African farmers, but can perhaps not look towards access to credit or decision-making. Agrivoltaic projects coupled with gender-responsive policies have the potential to optimize gains and bring more integrated social transformation (FAO, 2021).

Alternative Design and Outcomes

The adaptability of agrivoltaic system design is perhaps one of its strongest points, which allows for flexibility to accommodate diverse environmental conditions, types of crops, and community requirements. Fixed-tilt panels mounted above rows of crops are part of conventional designs, but new options have brought mobile tracking systems, vertical bifacial panels, and changeable canopy heights. All configurations have benefits as well as trade-offs (Barron-Gafford et al., 2019).

Vertical bifacial panels, for instance, are suitable in high-latitude regions or snow-dominated regions, where albedo effects can optimize sun absorption on both sides of the panel. These

systems minimize ground shadowing and facilitate the easy passage of traditional agricultural equipment. Dynamic systems involving the incorporation of sun tracking are most effective in terms of optimizing energy output, but are more costly to install and maintain and thus are less cost-effective in poor communities (Randle-Boggis et al., 2024).

Partially protected from the sun in arid and semi-arid regions like northern Kenya or parts of the Sahel, shade-tolerant crops like lettuce, chard, and tomatoes have been able to thrive. Studies have shown increased storage of water in shaded soils and reduced plant heat stress (Willockx et al., 2023). In addition, integrating agrivoltaics into livestock husbandry, through the use of raised installations that enable grazing underneath, delivers an integrated solution in agro-pastoral systems, achieving maximum food and energy productivity without sacrificing land area (IEA-PVPS, 2023).

Outcomes are made to vary and are based on community needs. Low-cost fixed panels with high-value crops have offered economic models for Indian smallholders (Zotti et al., 2024). On the other hand, in Japan, extremely sophisticated technology allows precision farming techniques and computerized irrigation systems to significantly reduce labor needs (Randle-Boggis et al., 2024).

Advantages, Risks, and Long-Term Implications

Agrivoltaics also has several advantages, which are in direct alignment with the United Nations' sustainable development goals. Land use efficiency, a term first coined, is precisely what agrivoltaics achieves: food and energy generation simultaneously on one hectare, double the quantity of yield per hectare a remedy most sought after in high-density or scarce-arable-land areas (Barron-Gafford et al., 2019). Second, the microclimate beneath panels also shields crops from extreme weather conditions like hail or exposure to too much sunlight, hence enhancing climate resilience.

Economically, agrivoltaics gives the farmer more than one source of income. Farm production on-farm can off-take power or farm income to manage farm price risk. Kenyan farmers producing for solar-irrigation cooperatives had up to 40% net improved farm income within two years (Willockx et al., 2023). In the long run, it can bring rural stability and reduce migration.

There are risks as well. Exorbitant capital costs, inadequate funding provision, and technical installation issues limit access. Inadequately thought-out schemes have unforeseen environmental side effects, such as pollinator corridors being severed or hydrology in the natural system being compromised (Randle-Boggis et al., 2024). In addition, unless they are well-controlled in an equilibrated way, technology will serve to cement inequality. More and more, massive commercial agrivoltaic farms are also increasingly displacing smallholders from their lands, placing the ownership of lands in the hands of energy multinationals or overseas investors (FAO, 2021).

Lastly, long-term to medium-term successful agrivoltaic integration requires systems thinking to accompany policy facilitation, inclusive finance instruments, and ongoing capacity building.

Conclusion

Agrivoltaics offers a vision of a sustainable future through synchronizing food and energy requirements, as well as an integrated land policy. From Japan's high-tech systems high tech to Kenya's cooperative societies, agrivoltaics has been shown to have the ability to reimagine rural livelihoods, increase climate resilience, and reduce environmental damages. Do it on a large scale, however, particularly in Africa, and it requires more than technology to achieve. Radical transformation of the policy systems, finance systems, education systems, and land administration systems is needed.

African countries, despite all their vulnerability and promise, can gain much from agrivoltaics if the strategy is wise.

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